Note 2 Solutions

Exercise 1: Generalization of Theorem 2.2

Claim: For any positive integer n, if the sum of the digits of n is divisible by 9, then n is divisible by 9.

Proof. Let n be a positive integer with k digits, and write a_i for the i-th digit of n, so that

$$n = \sum_{i=0}^{k-1} a_i \cdot 10^i.$$

Assume that the sum of the digits of n is divisible by 9, i.e., $\exists\, m \in \mathbb{Z}$ such that

$$\sum_{i=0}^{k-1} a_i = 9m. \tag{1}$$

We use the identity $10^i = (10^i - 1) + 1$ to rewrite n as:

$$n = \sum_{i=0}^{k-1} a_i \cdot 10^i = \sum_{i=0}^{k-1} a_i(10^i - 1) + \sum_{i=0}^{k-1} a_i.$$

By (1), the second sum equals $9m$. For the first sum, note that for each $i \geq 0$,

$$10^i - 1 = \underbrace{99\cdots9}_{i \text{ nines}},$$

which is divisible by 9. Therefore $9 \mid a_i(10^i - 1)$ for each i, so $9 \mid \sum_{i=0}^{k-1} a_i(10^i - 1)$.

Hence both terms are divisible by 9, and we conclude $9 \mid n$. □

Exercise 2: Proof of Lemma 2.2

Lemma 1. *If a^2 is even, then a is even.*

Direct Proof

Proof. Assume a^2 is even. Since a is an integer, it is either even or odd.

- If a is even, we are done.
- If a is odd, then $a = 2m + 1$ for some $m \in \mathbb{Z}$, so

  $$a^2 = (2m+1)^2 = 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1,$$

  which is odd. This contradicts our assumption that a^2 is even.

Therefore a must be even. □

Proof by Contraposition

Proof. The contrapositive of the statement is: if a is odd, then a^2 is odd.

Assume a is odd. Then $a = 2m + 1$ for some $m \in \mathbb{Z}$. Squaring:

$$a^2 = (2m+1)^2 = 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1,$$

which is odd. Therefore a^2 is odd, and the contrapositive is proved. □

Since the contrapositive is logically equivalent to the original statement, Lemma 2.2 follows. □

Which proof is better suited?

The proof by contraposition is better suited to this lemma. In the contrapositive direction, we assume a is odd and work *forward* via straightforward algebra to conclude a^2 is odd. The direct proof, by contrast, requires reasoning *backwards* from a property of a^2 to a property of a, which forces us to introduce a case analysis and derive a contradiction. Contraposition avoids this by letting us work in the easier direction.